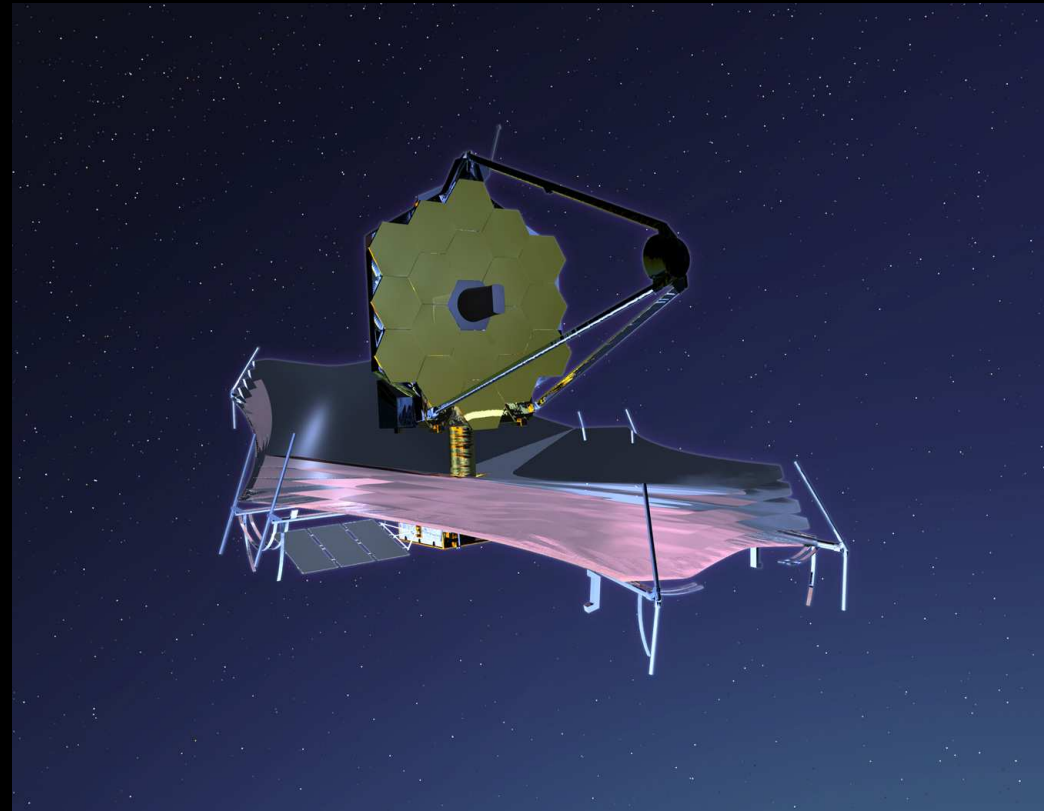


Using Hubble Object Finding and Classification Software to Measure Cancer Cell Spreading and Diabetes Type 2 Markers in an Early Stage

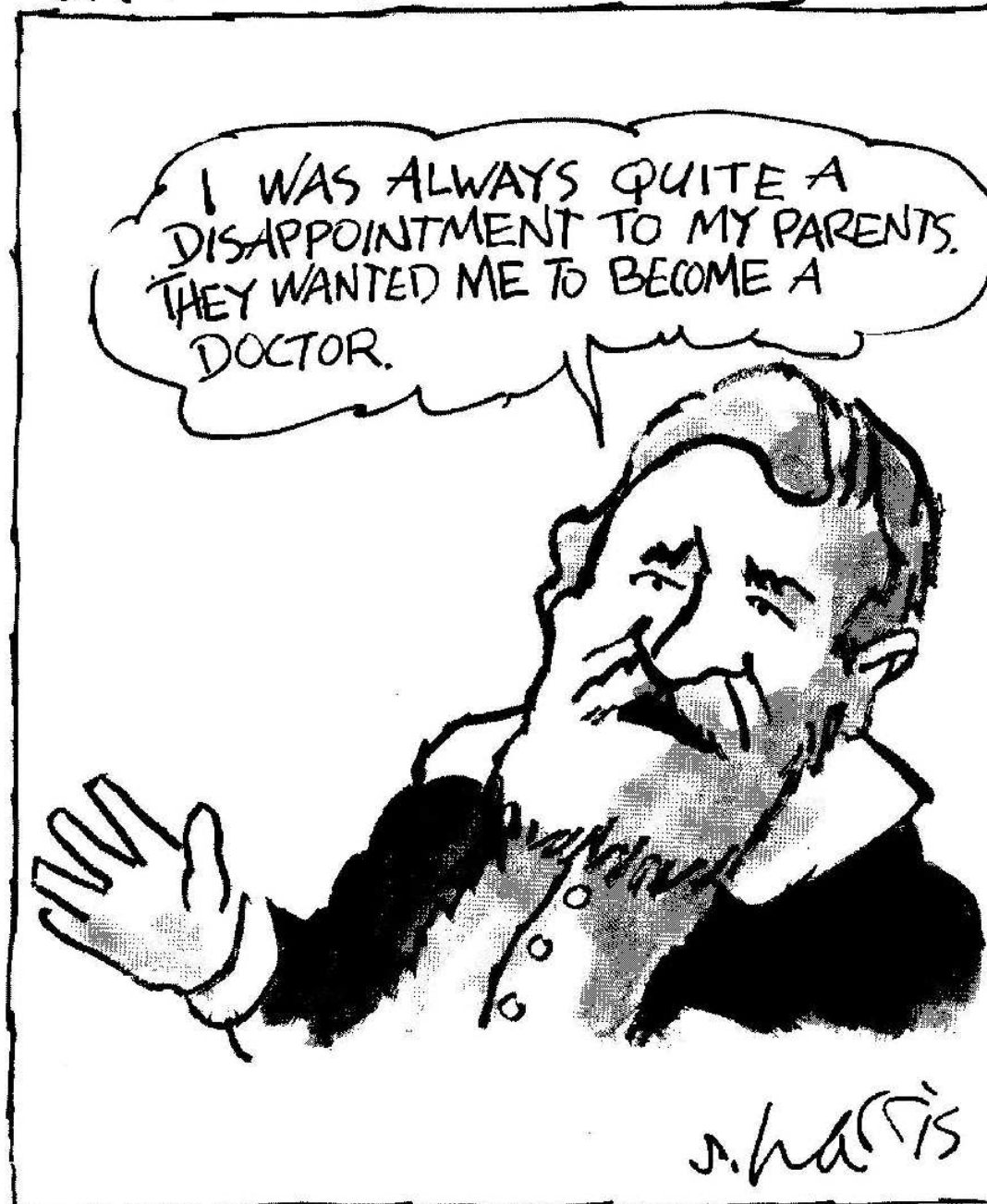
Rogier Windhorst (School of Earth and Space Exploration, ASU)

P. Baluch, L. Burnett, J. Brower, A. Casano, J. Olson (ASU), R. & J. Herman (Banner Health), R. Ryan, K. Tamura, V. Taylor (ASU)



Research Summary to Aperio, from ASU Skysong, Thursday Sept. 23, 2010, Tempe, AZ

GALILEO'S BURDEN



Why I personally got involved in medical research ... (true story!)

Outline

(1) Summary of the NASA Hubble Space Telescope connection:

- (1a) NASA missions to study Cosmic Dawn, First Light, Dark Ages.
- (1b) NASA Hubble Space Telescope (HST) image analysis.

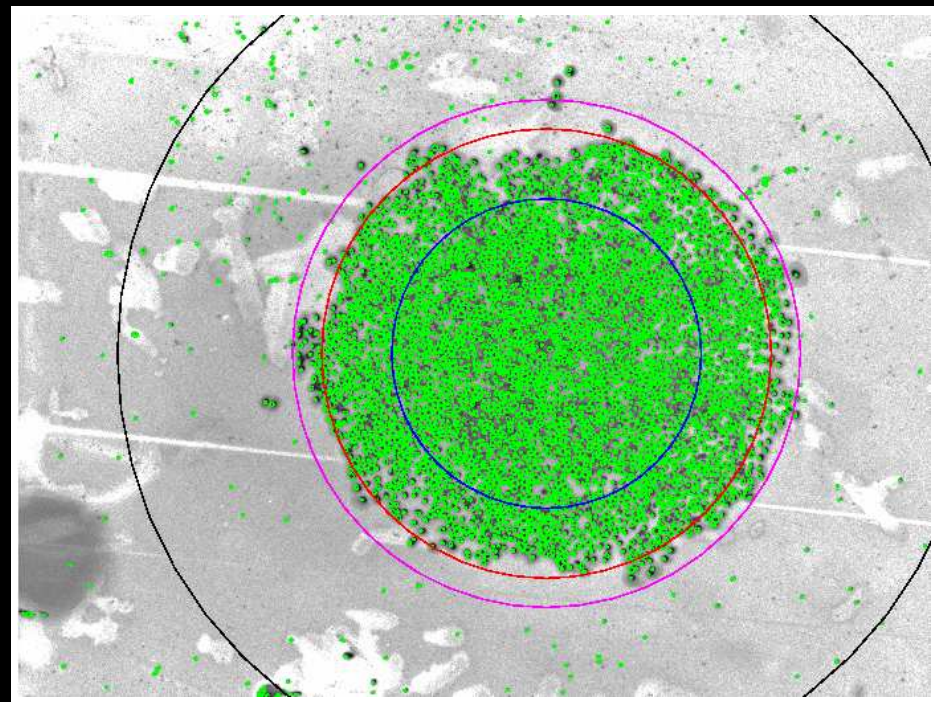
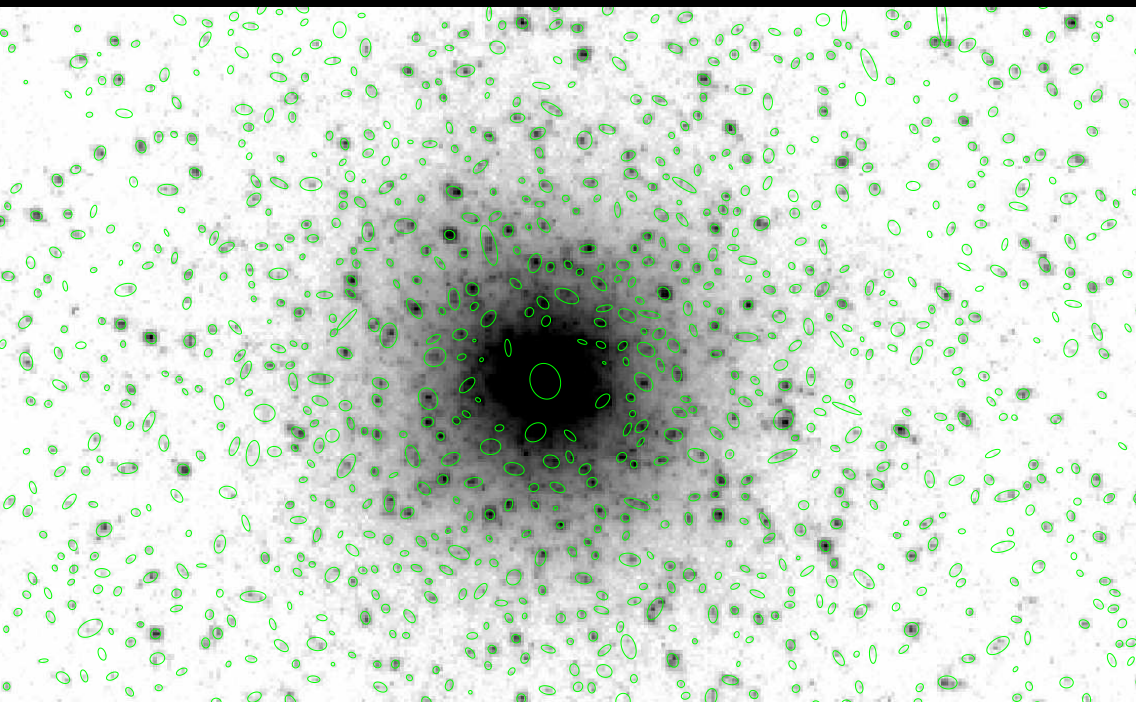
(2) Using Hubble Object Finding and Classification Software to Find Cancer Cell Spreading and Markers for Diabetes Type 2 in an Early Stage:

- (2a) The Object Finding, Detection, Deblending, and Classification method.
- (2b) Application of Hubble Software to Study Spreading of Cancer Cells.
- (2c) Application to C-fiber Skin-biopsies to Normal and Obese Patients.

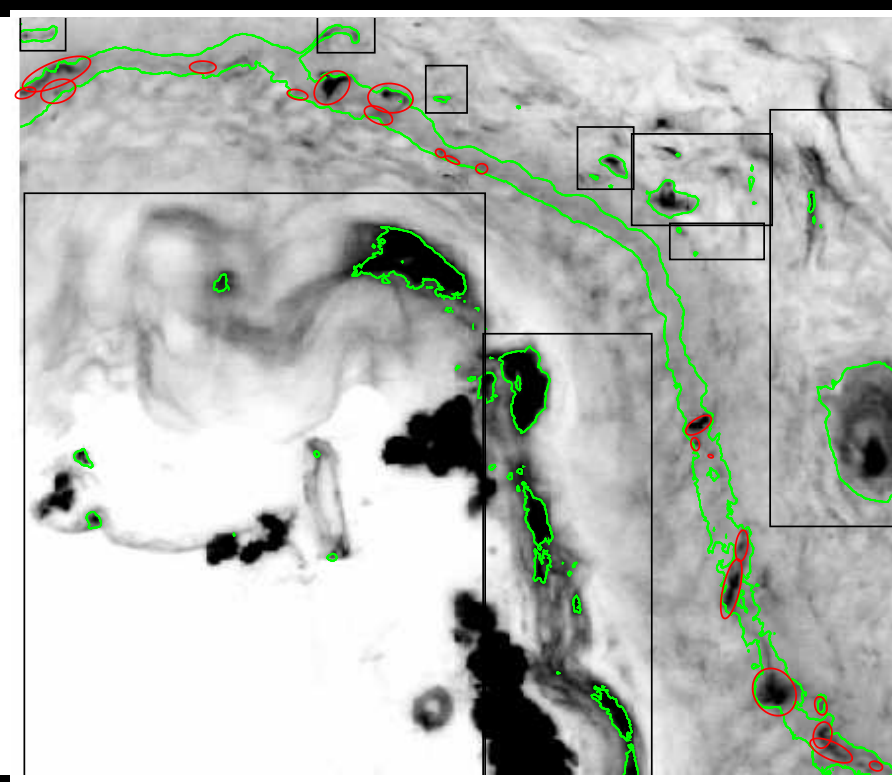
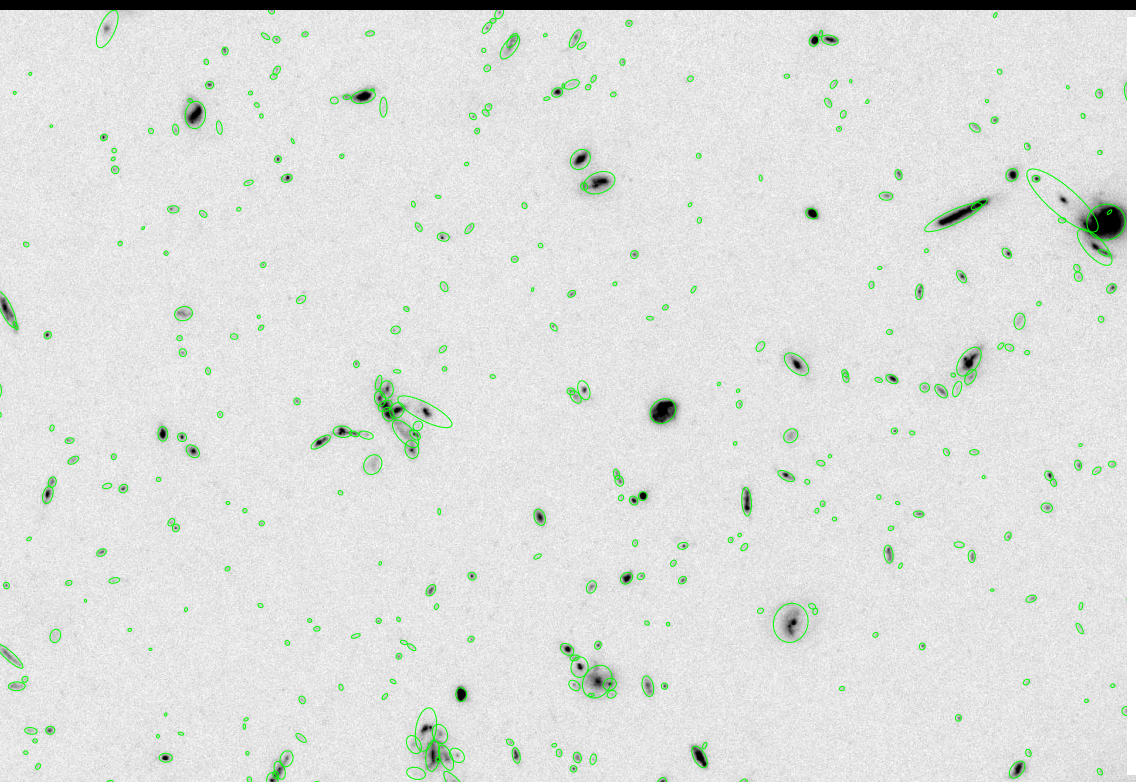
(3) Results and and Conclusions:

- Reliably find faint features in non-uniform, noisy background.
- Reliably process large numbers of images at once (msec of CPU time).

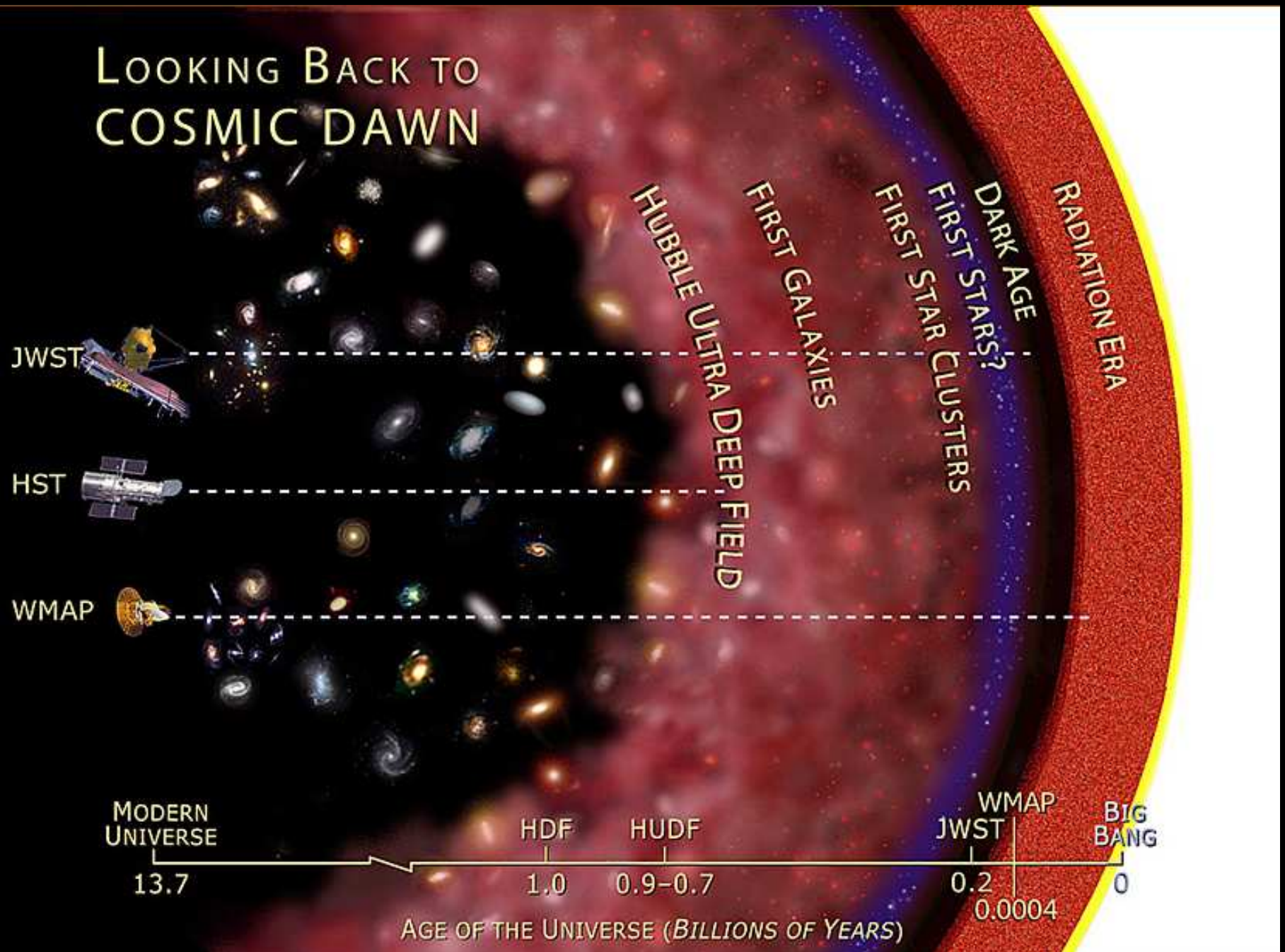
Sponsored by NASA/ASU/Banner Health



Hubble star-cluster & galaxies (left); Bladder cancer & C-fiber biopsies (right).

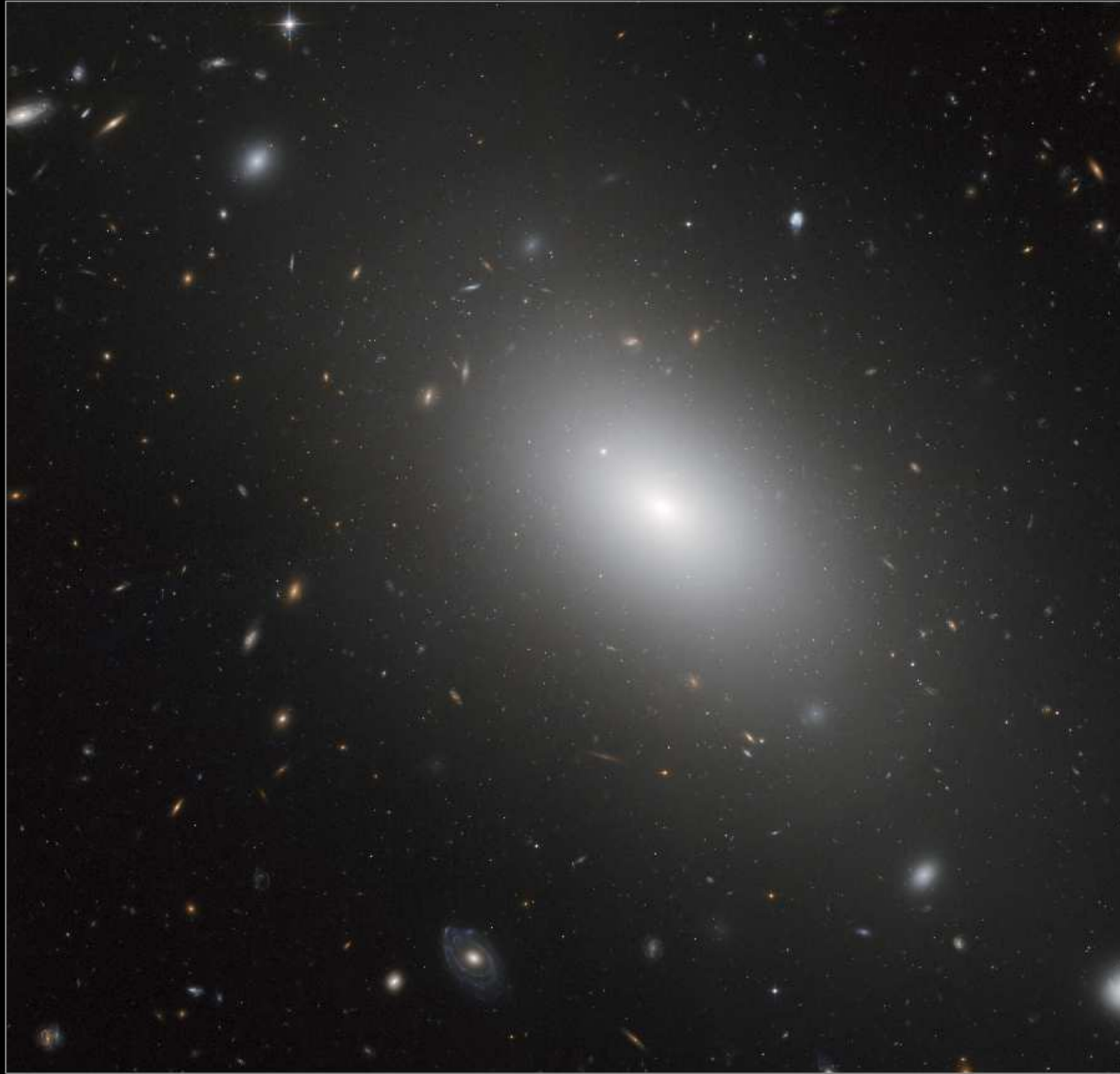


(1a) NASA missions that study Cosmic Dawn, First Light, & the Dark Ages



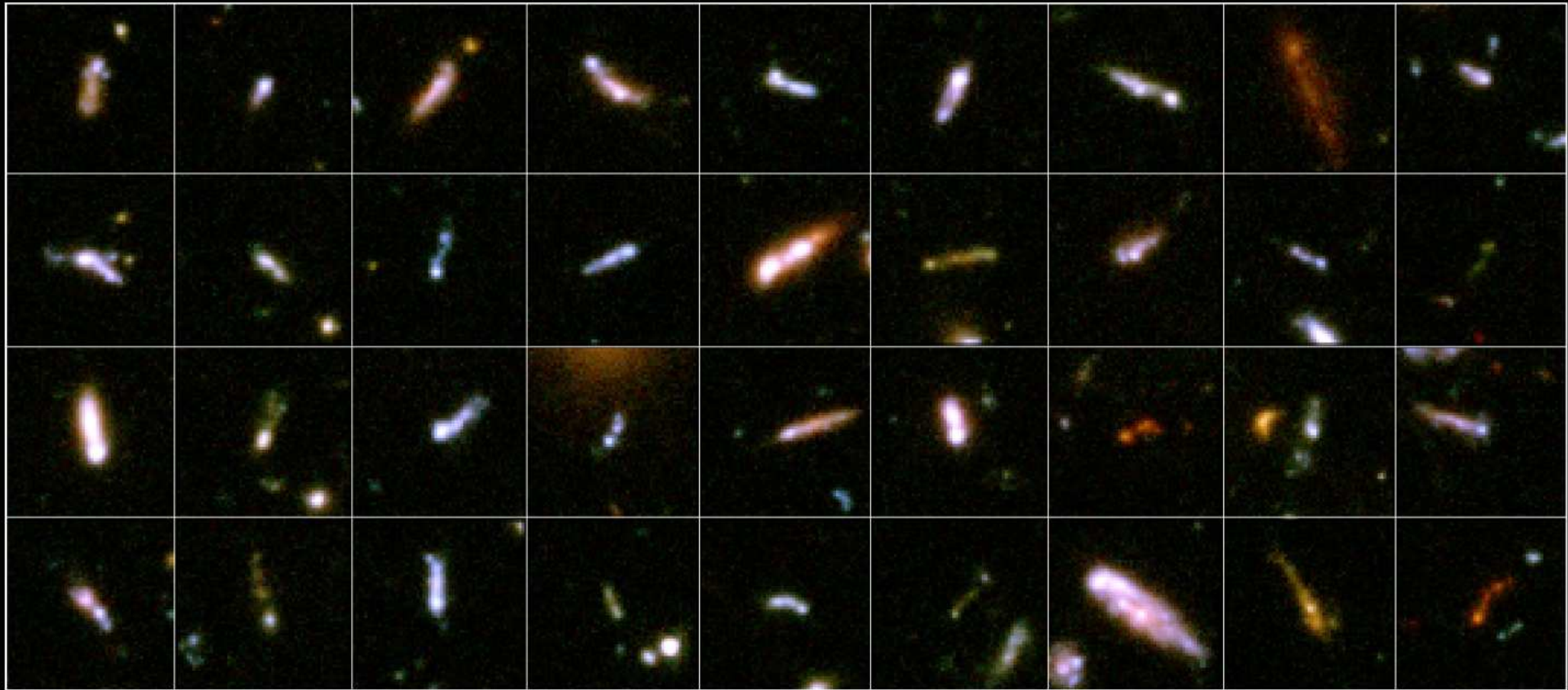
Looking back in time $\implies$ deal with many overlapping galaxy images!

Elliptical Galaxy NGC 1132



Hubble
Heritage





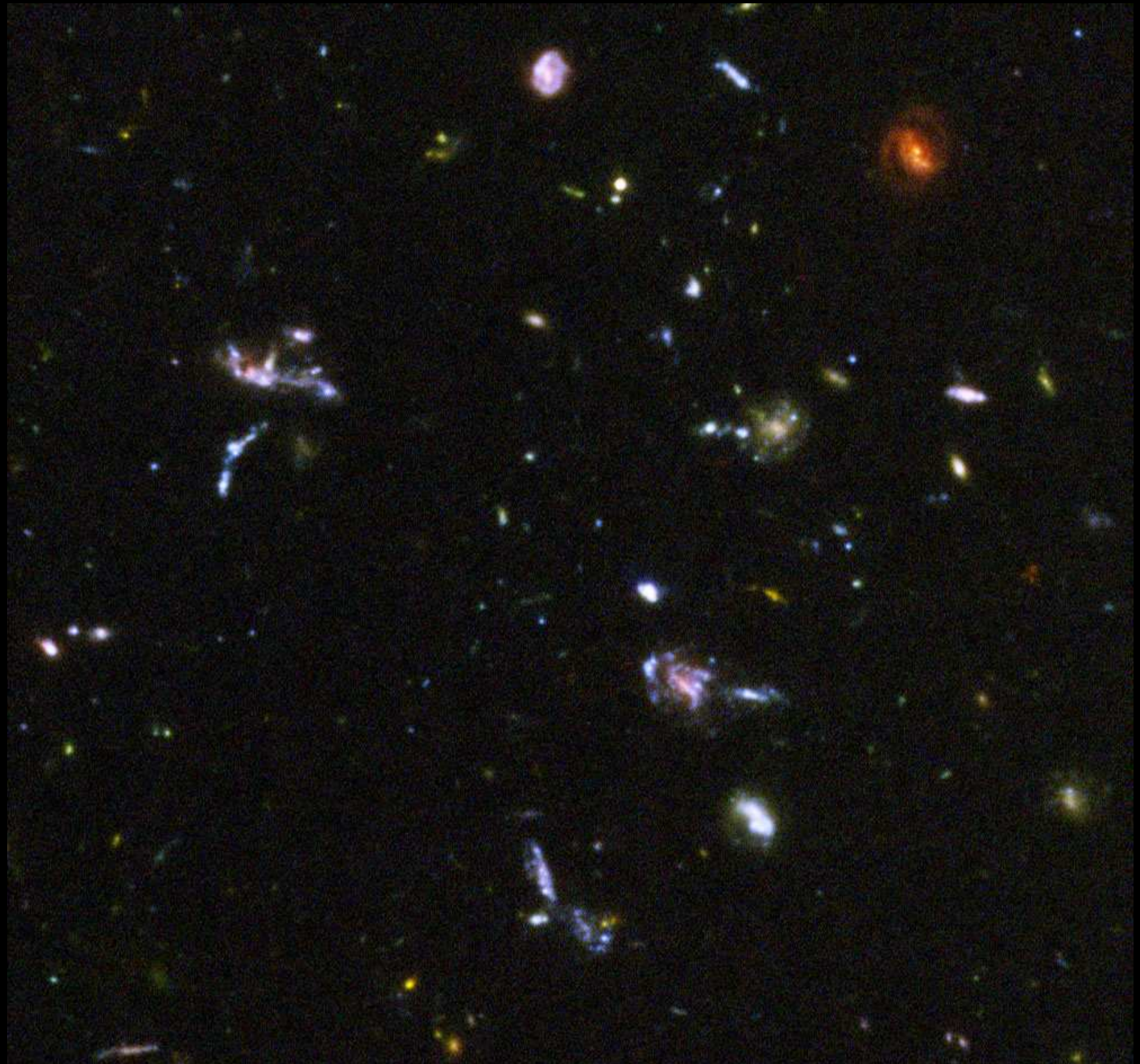
“Tadpole” Galaxies in the Hubble Ultra Deep Field
Hubble Space Telescope ■ ACS/WFC

NASA, ESA, A. Straughn, S. Cohen and R. Windhorst (Arizona State University), and the HUDF team (STScI)

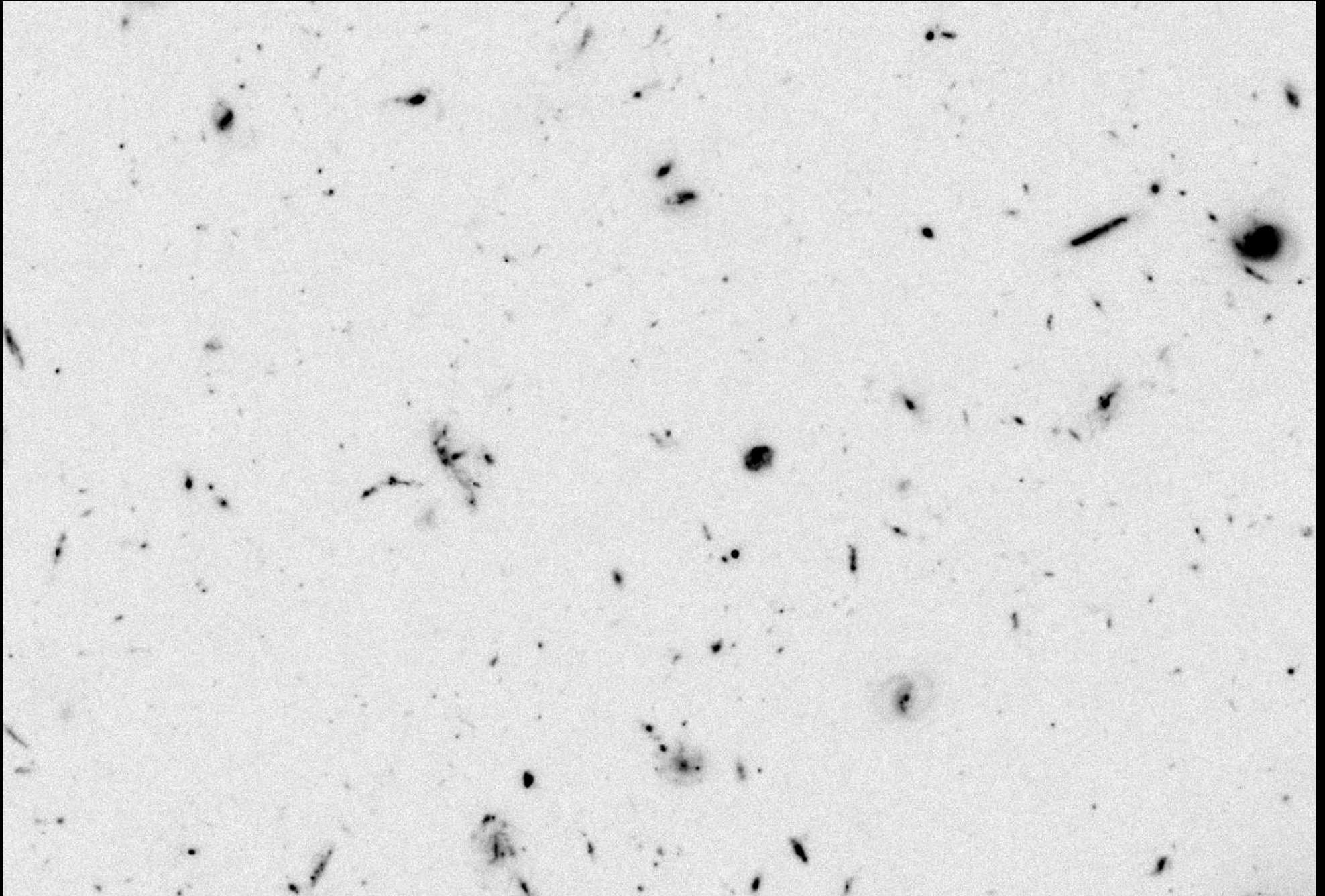
STScI-PRC06-04

Merging galaxies constitute $\lesssim 1\%$ of Hubble sequence today (age $\gtrsim 12.5$ Gyr).

Tadpole galaxies are early stage mergers, very common at $z \gtrsim 2$ (age $\lesssim 3$ Gyr).

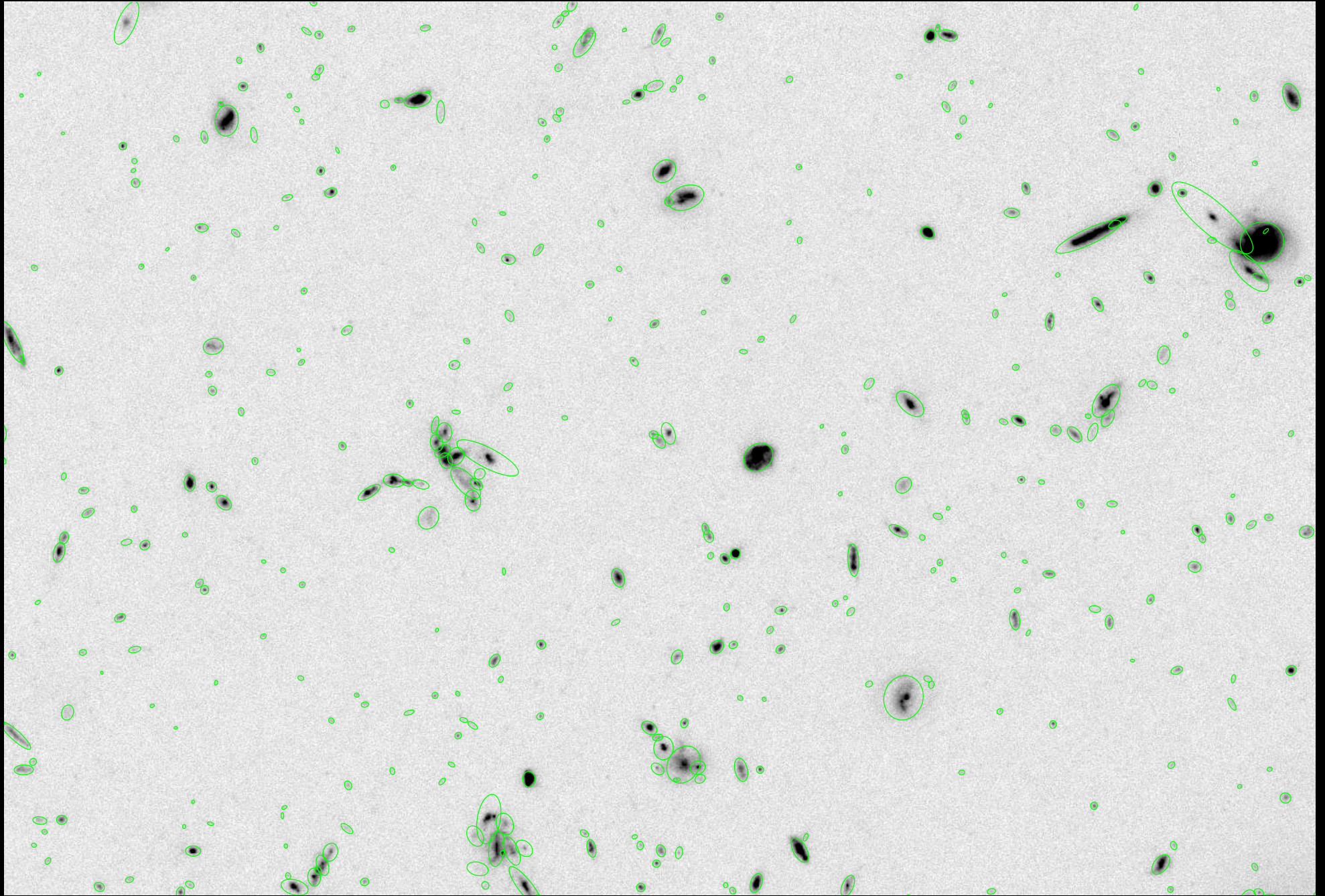


(2a) Object Finding, Detection, Deblending, and Classification method



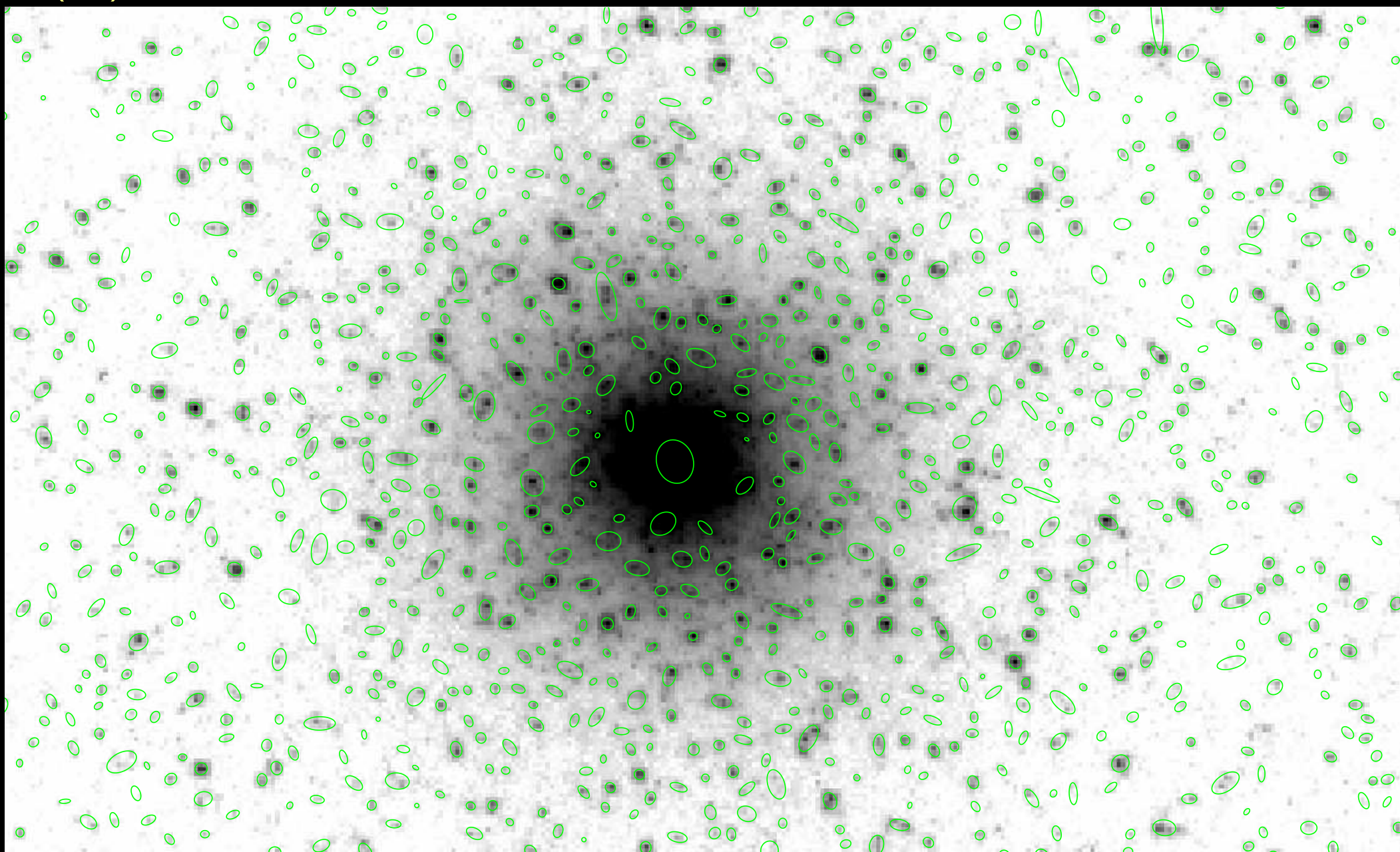
Very complex filamentary galaxies are seen in early stages of formation.

(2a) The Object Finding, Detection, Deblending, and Classification method



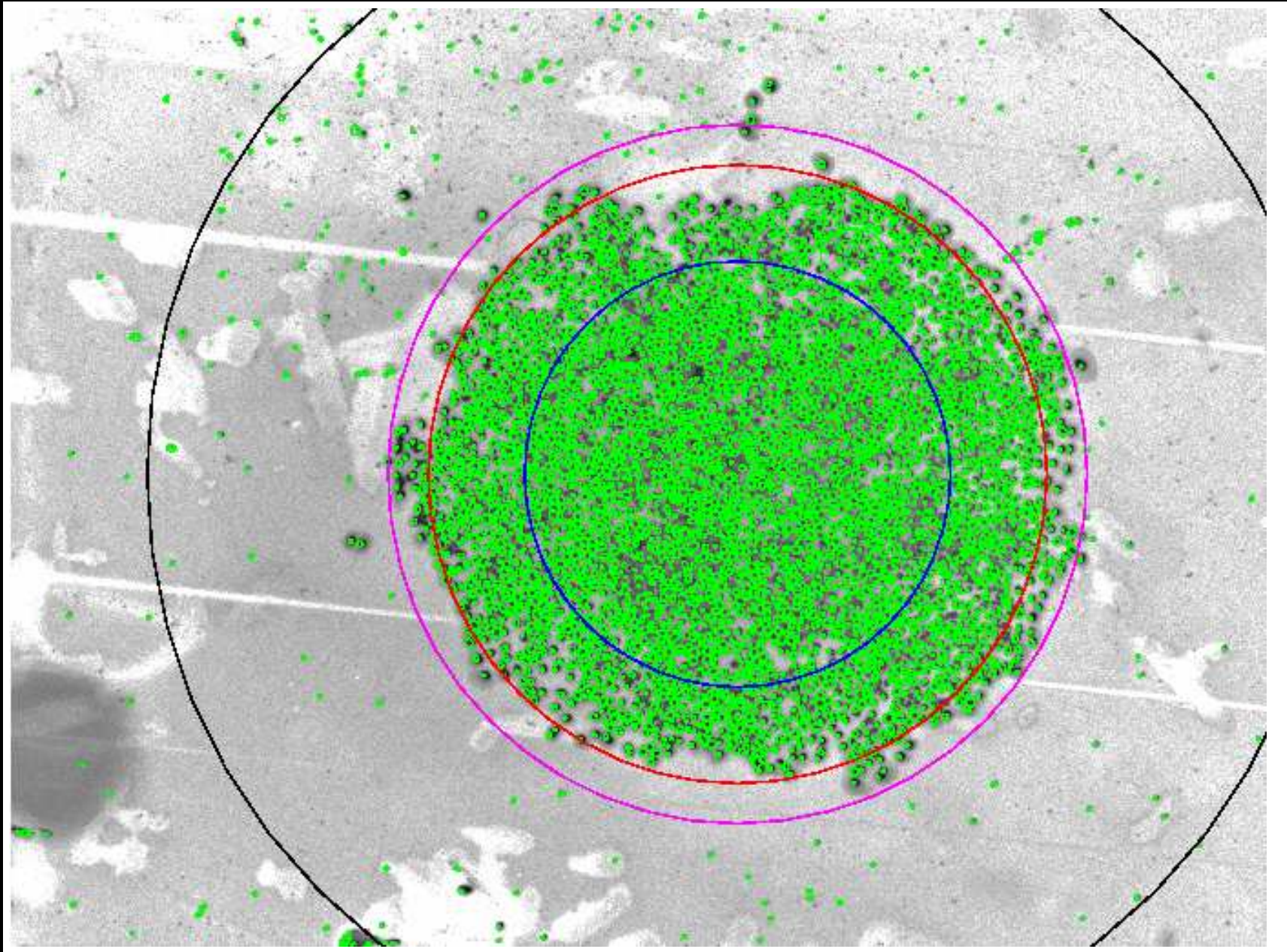
The Source-Extractor software deblends these in chains of elliptical isophotes.

(2b) The Object Finding, Detection, Deblending, and Classification method



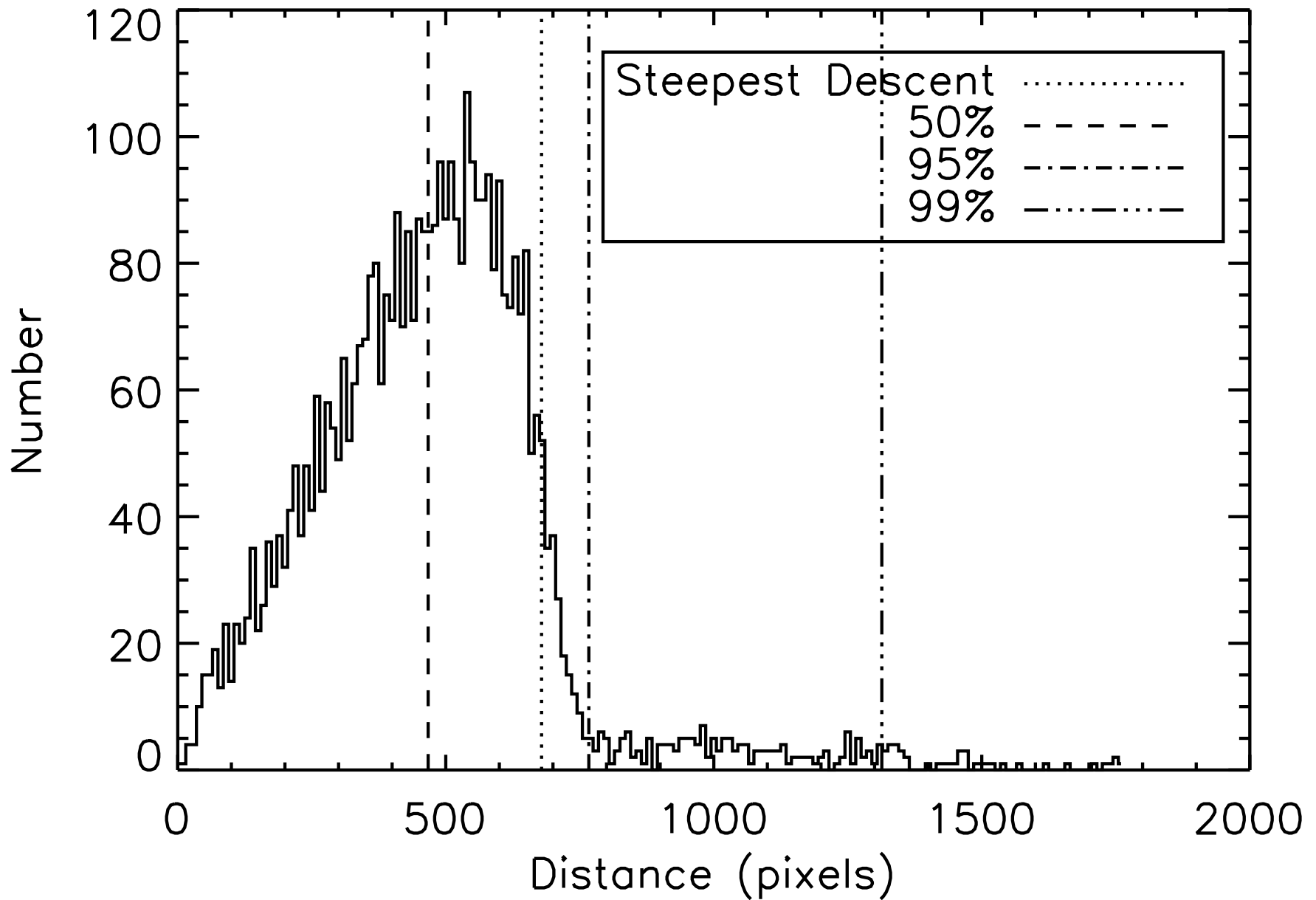
The Source-Extractor algorithm deblends a rich globular cluster of 10^5 stars.
(Only several 10^3 brightest stars plotted, but object overlap is allowed).

(2b) Application of Hubble Software to Study Spreading of Cancer Cells



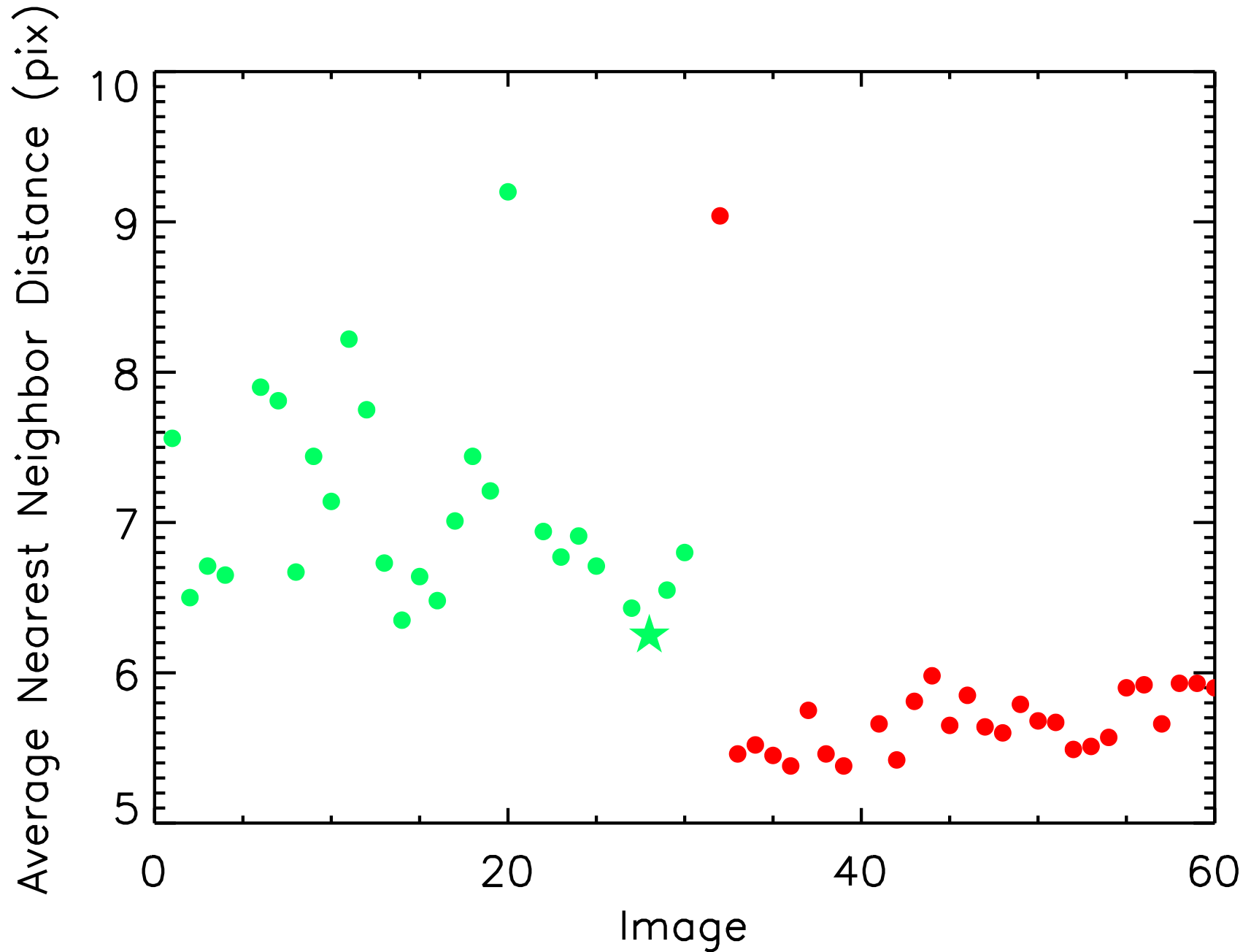
Source-Extractor directly counts number of cells in a cancer biopsy image.
(Collaboration with Dr. M. Berens et al. at TGEN, R. Ryan at ASU.)

(2b) Application of Hubble Software to Study Spreading of Cancer Cells



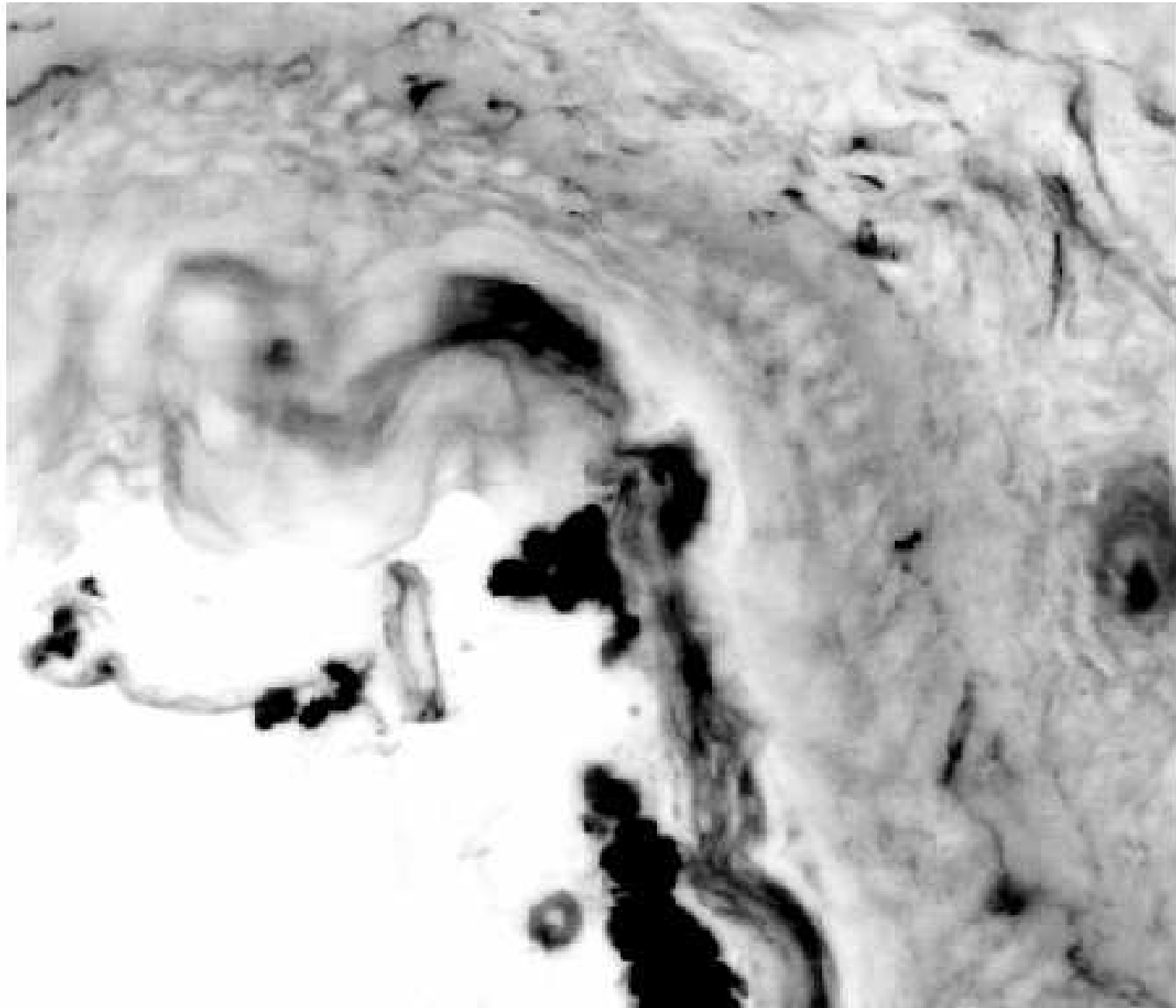
The distribution and spreading of cancer cells is measured quantitatively.

(2b) Application of Hubble Software to Study Spreading of Cancer Cells



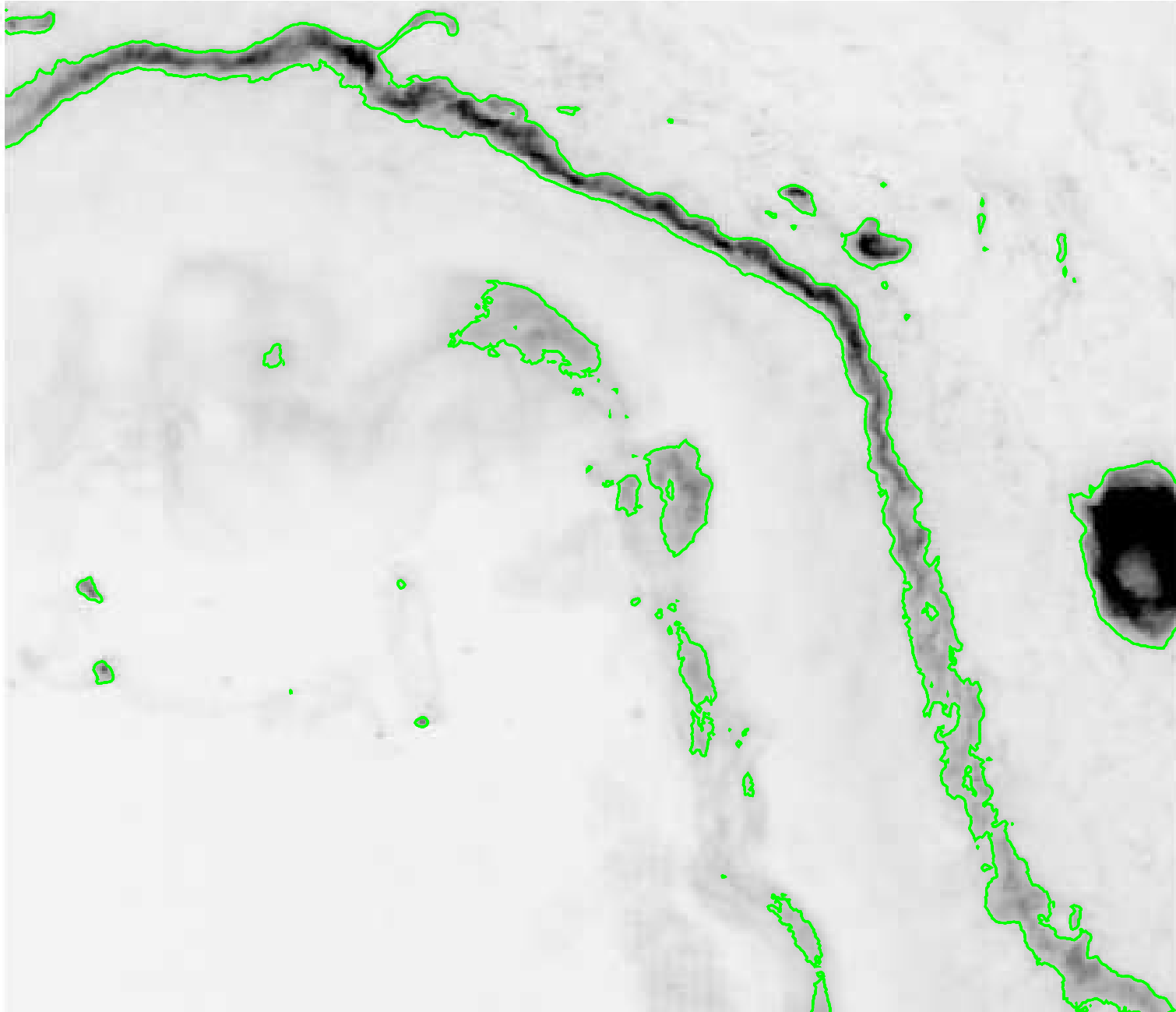
Nearest neighbor analysis bifurcates the spread and non-spread cancer biopsies.

(2c) Application to C-fiber Skin-biopsies in Normal and Obese Patients



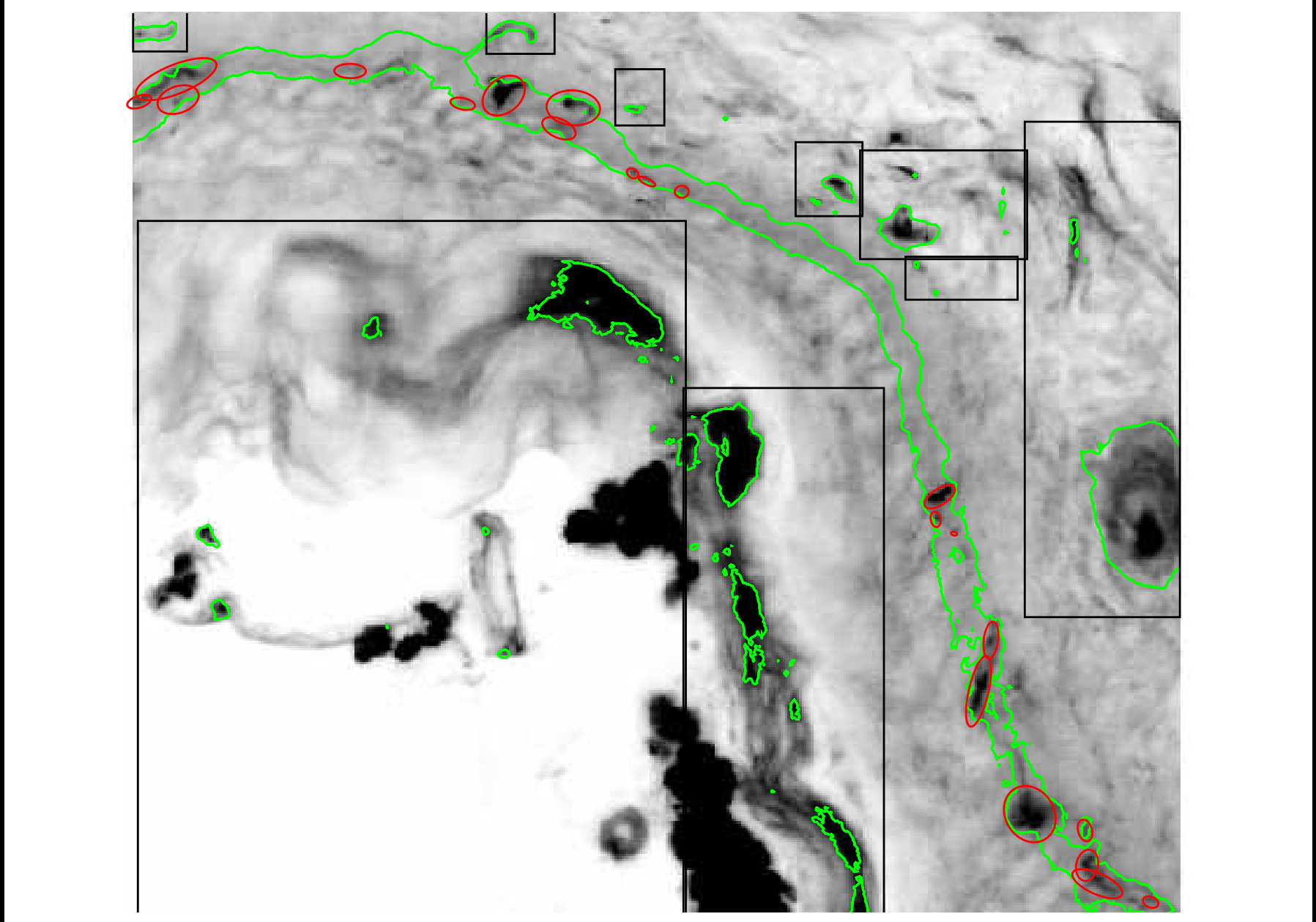
A Stained Confocal Microscope image of a Skin-Biopsy is digitized.

(2c) Application to C-fiber Skin-biopsies in Normal and Obese Patients



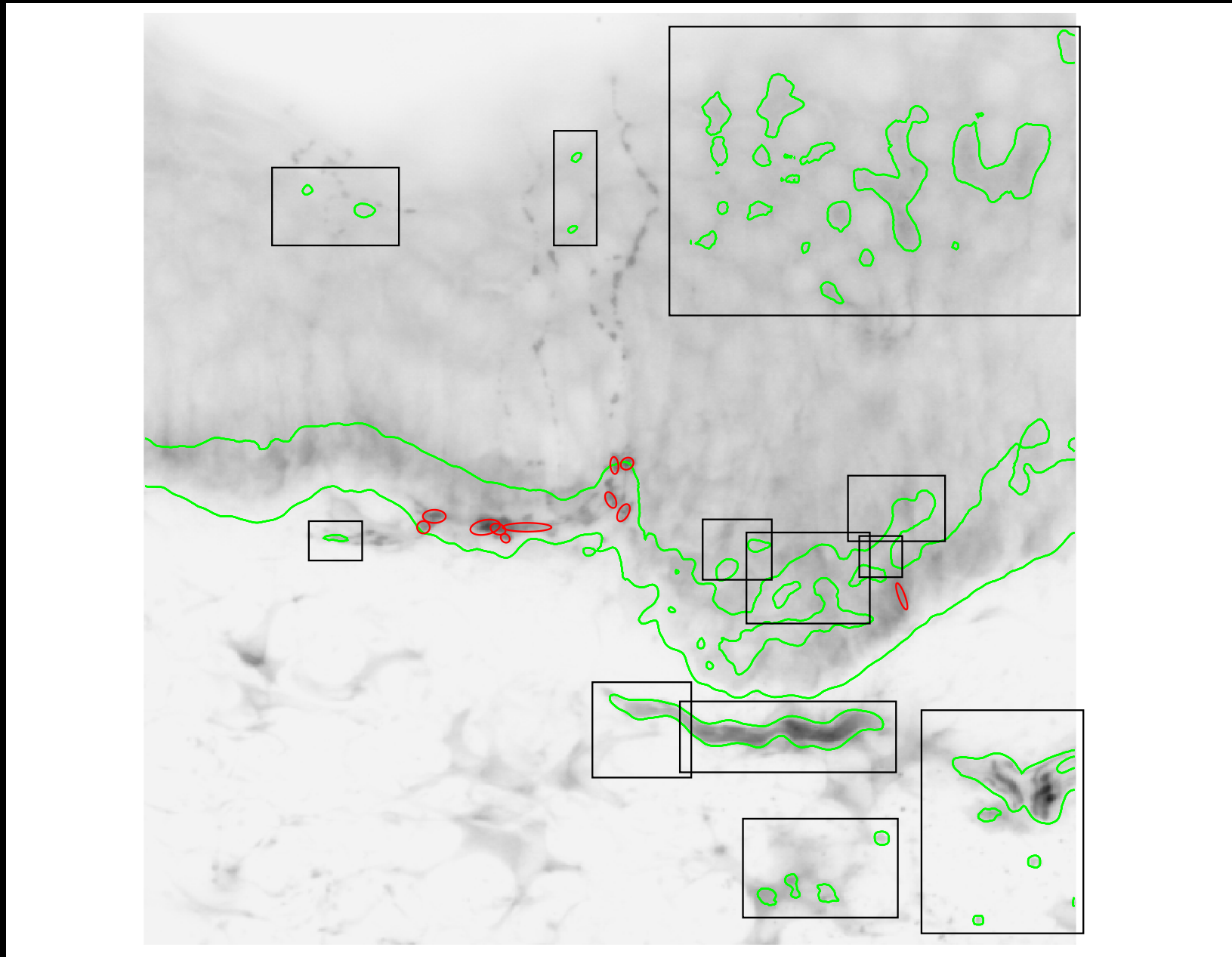
Different staining allows us to define the membrane and other parts of skin.

(2c) Application to C-fiber Skin-biopsies in Normal and Obese Patients



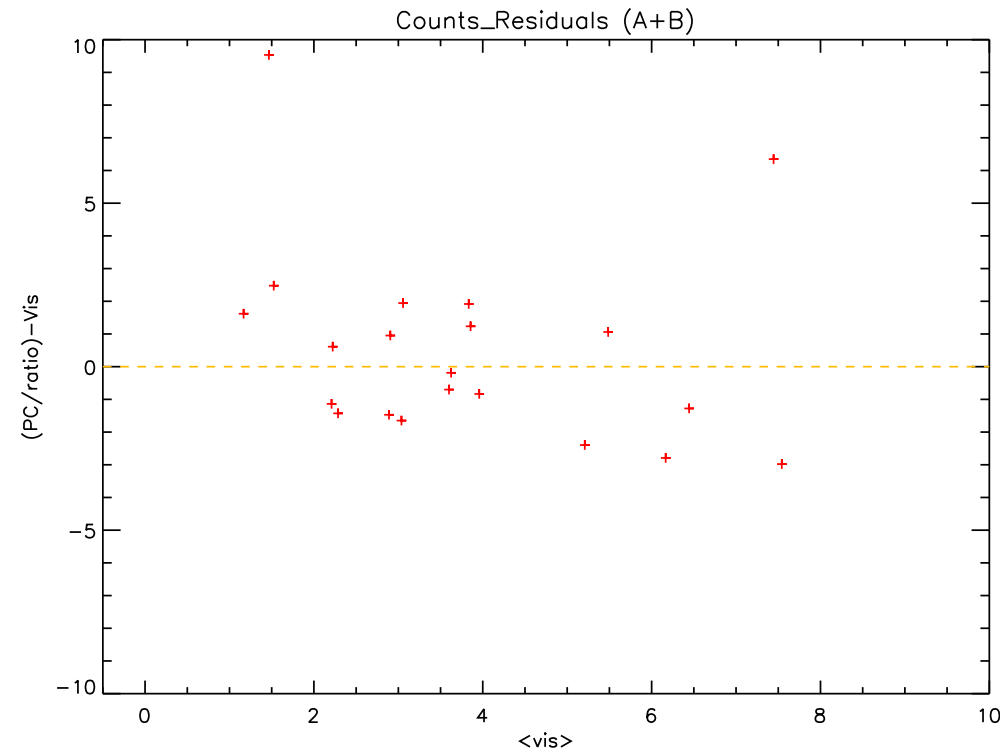
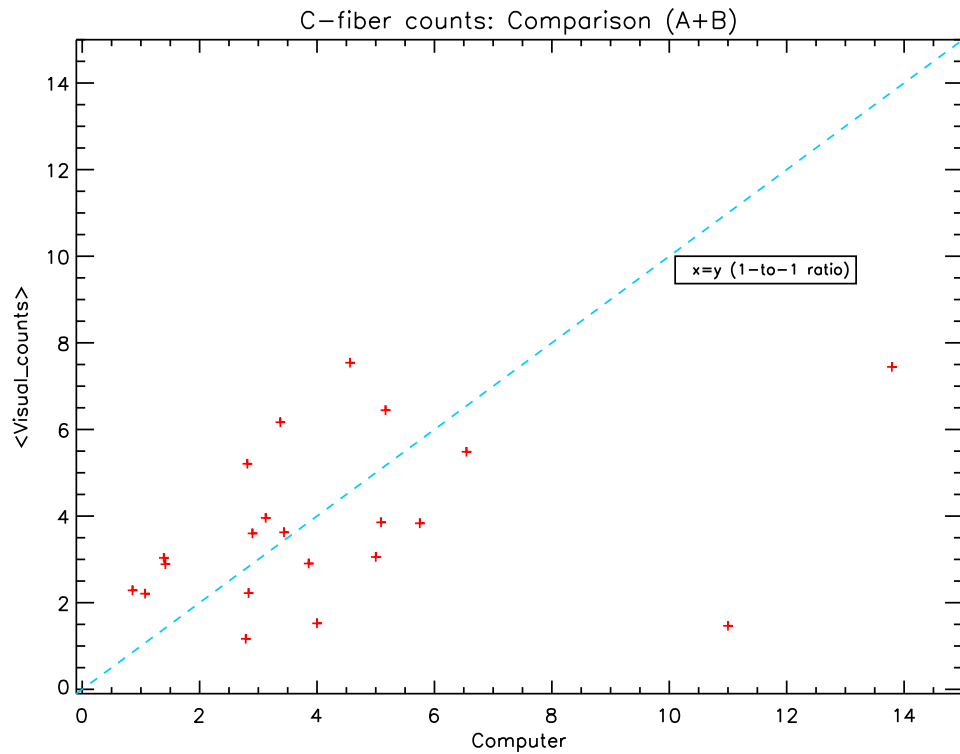
The Source-Extractor finds C-fibers that cross the membrane (in red).
Black boxes: Exclusion zone where Source-Extractor avoids bogus objects.

(2c) Application to C-fiber Skin-biopsies in Normal and Obese Patients



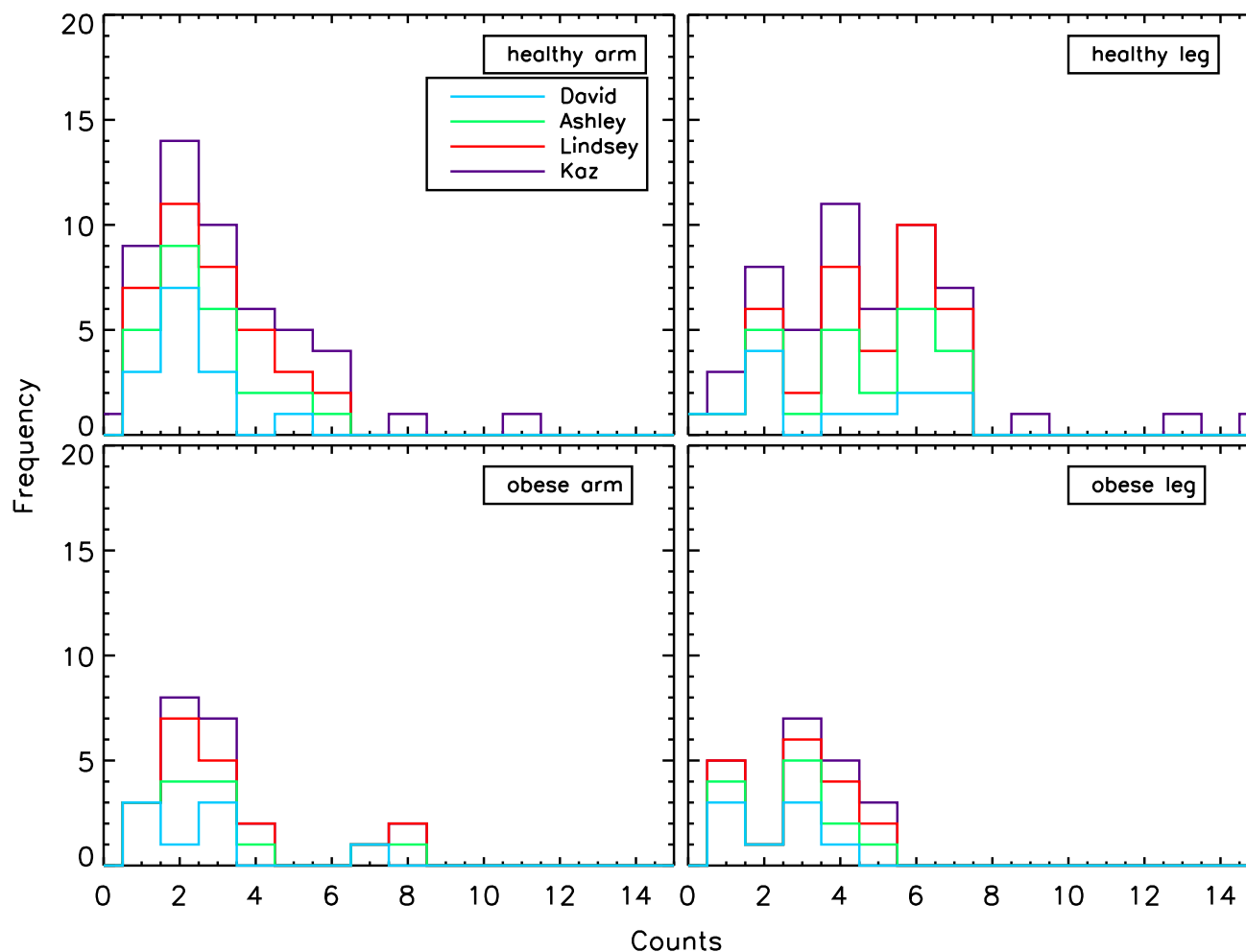
The Source-Extractor finds C-fibers that cross the membrane (in red).
Black boxes: Exclusion zone where Source-Extractor avoids bogus objects.

(2c) Application to C-fiber Skin-biopsies



- Machine count correlates well with visual counts, with a few catastrophic failures where software breaks a C-fiber up in too many pieces.
- Average fragmentation is calibrated from expert visual classifiers.
- Catastrophic failures are corrected by hand — working on training the software to recognize and correct those as well.
- Dispersion between machine and visual classifiers is $\sigma \simeq \pm 1.5$ count.

(2c) Application to C-fiber Skin-biopsies



- Cumulative counts of both the machine and the visual classifiers show different distributions for normal and obese patients.
- Obese patients have in general lower C-fiber counts, although the differences between the two populations are not always huge.

(3) Results and Conclusions

(1) Summary of the NASA Hubble Space Telescope connection

- (1a) NASA Hubble Space Telescope image analysis software can detect, deblend and classify overlapping faint galaxy or faint star images.
- (1b) Software is very general — doesn't care what it counts or classifies!

(2) Can use software to identify spreading of Cancer cells or potential markers for Diabetes Type 2 in early stage:

(2a) Galaxy Detection, Deblending, and Classification software is fairly robust and general, minimizing human interaction.

(2b) Application of Hubble Software also allows us to quantitatively study the spreading of Cancer Cells in biopsies.

(2c) Application to C-fiber Skin-biopsies in Normal and Obese patients allows us to quantitatively distinguish healthy from sick patients.

- Reliably find faint features in non-uniform, noisy background.
- Reliably process large numbers of images at once (msec of CPU time).

REFERENCES

<http://hubblesite.org/newscenter/2010/01/>

Casano, A. R., Brower, J. B., Baluch, D. P., Windhorst, R. A., Targovnik, J. H., & Herman, R. M. 2009, *International Journal of Obesity*, submitted (Jul. 2010) [<http://www.nature.com/ijo/index.html>] “Small Sensory Nerve Fiber Neuropathy in Morbidly Obese Subjects: Reduced Intra Epidermal Nerve Fiber Density”

Herman, R. M., Brower, J. B., Stoddard, D. G., Casano, A. R., Targovnik, J. H., Herman, J. H., & Tarse, P. 2006, *International Journal of Obesity*, 31, 226–235 [www.nature.com/ijo/journal/v31/n2/full/0803418a.html] “Prevalence of somatic small fiber neuropathy in obesity ”

Tamura, K., Mager, V. A., Burnett, L. A., Olson, J. H., Brower, J. B., Casano, A. R., Baluch, D. P., Targovnik, J. H., Windhorst, R. A., & Herman, R. M. 2009, *Journal of Neuroscience Methods*, 185, pg. 325–337 [<http://ees.elsevier.com/jneumeth/>] “Semi-Automated Method of Analysis of Small Sensory Nerve Fibers in Human Skin-Biopsies”

SPARE CHARTS

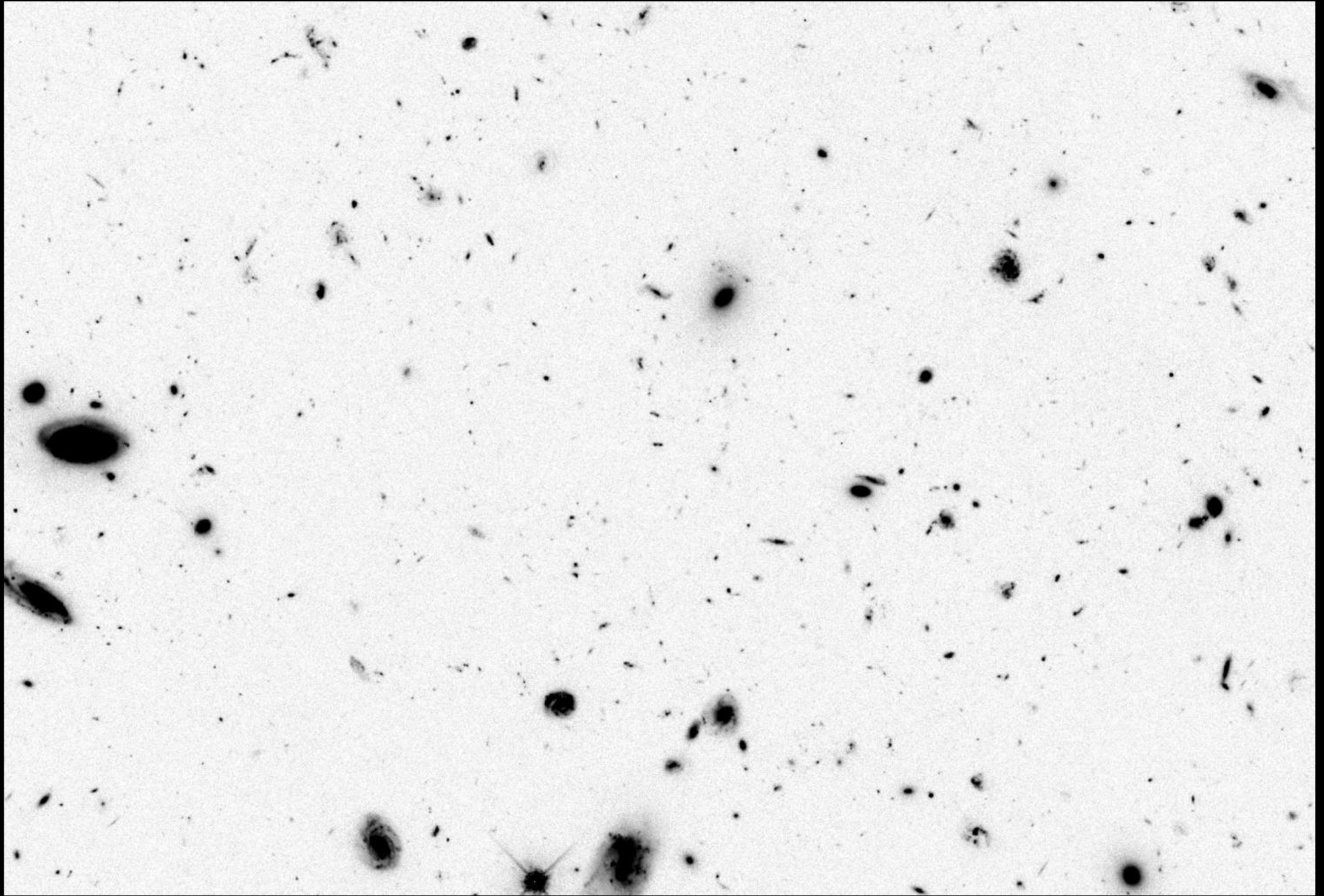
(1) The NASA Hubble Space Telescope connection

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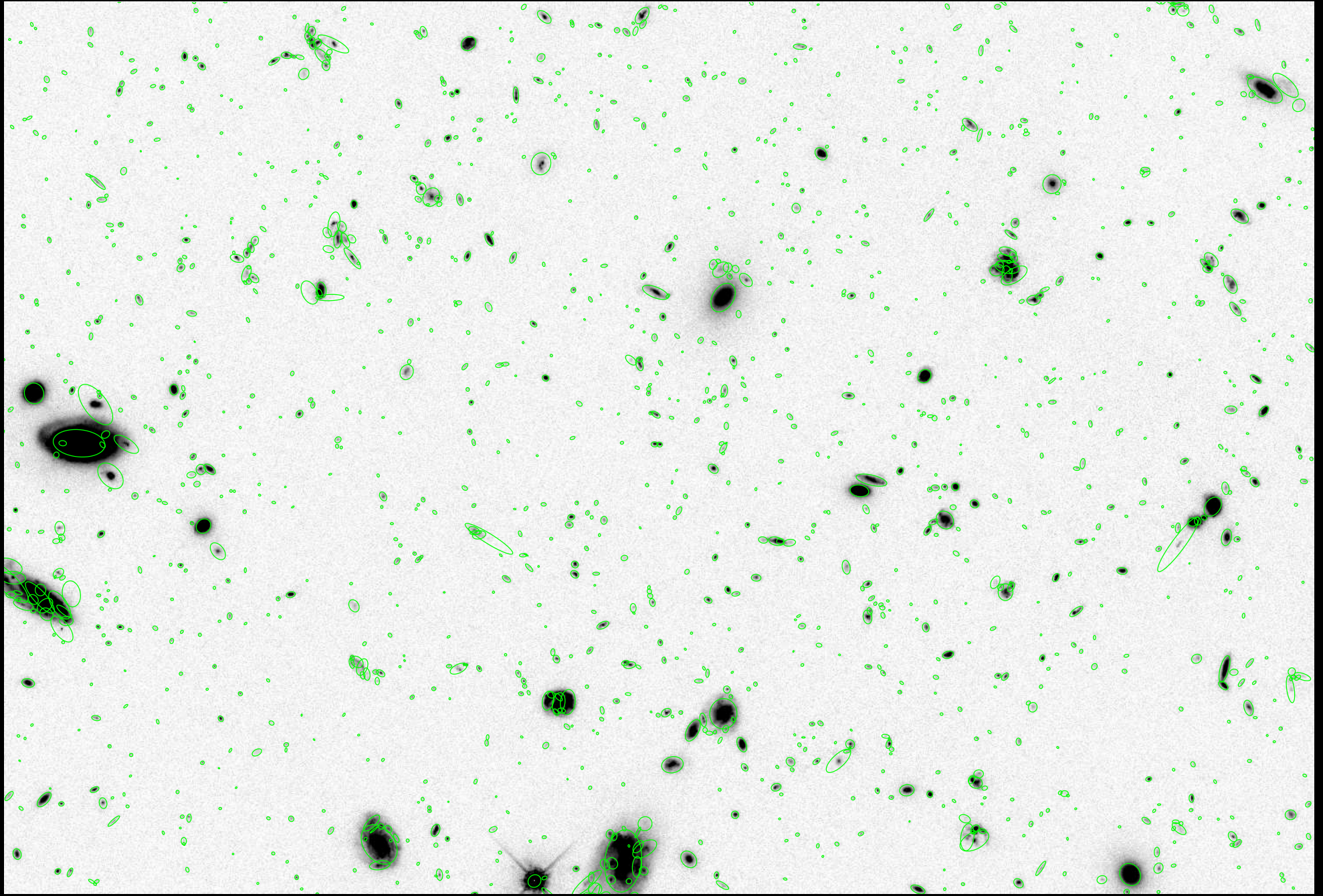
This project of medical applications of Hubble galaxy classification software builds on a 40 year history of astronomical software development.

(1b) NASA Hubble Space Telescope image analysis

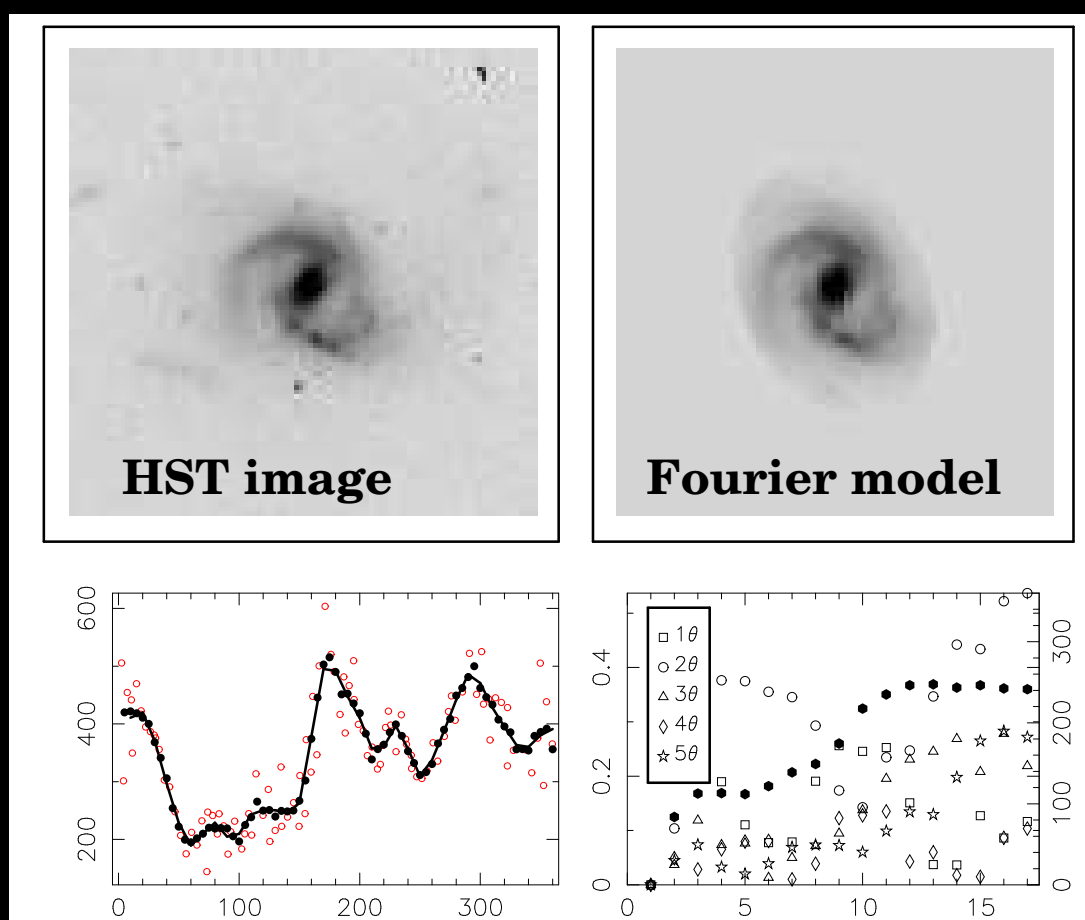


The deepest Hubble images start showing significant galaxy overlap.

(1b) NASA Hubble Space Telescope image analysis



to address crowding, the source-extractor code iteratively deblends the images.



Fourier Decomposition is a robust way to measure galaxy morphology and structure in a quantitative way (Odewahn et al. 2002):

- (1) Fourier series are made in successive concentric annuli.
- (2) Even Fourier components indicate symmetric parts (arms, rings, bars).
- (3) Odd Fourier components indicate asymmetric parts (lopsidedness).
- (4) JWST can measure the evolution of each feature/class directly.